```
#import the csv file
from google.colab import files
uploaded = files.upload()
```

Choose Files sample_cof…les_100.csv

sample_coffee_shop_sales_100.csv(text/csv) - 3687 bytes, last modified: 10/26/2026 - 100% done

```
Saving sample_coffee_shop_sales_100.csv to sample_coffee_shop_sales_100.csv
```

```
import numpy as np
import pandas as pd

df = pd.read_csv("sample_coffee_shop_sales_100.csv")

df.head()
```

	Date	Product	Units_Sold	Unit_Price	Discount_Percent	Customers	Revenue
0	2026-01-01	Juice	17	100	5	26	1615.0
1	2026-01-02	Tea	20	60	5	52	1140.0
2	2026-01-03	Tea	40	60	5	98	2280.0
3	2026-01-04	Muffin	12	80	0	44	960.0
4	2026-01-05	Muffin	14	80	0	30	1120.0

Next steps: Generate code with df | New interactive sheet

```
#Creating numpy array and performing operations
revenue_array = np.array(df["Revenue"])

print("Total Revenue:", np.sum(revenue_array))
print("Minimum Revenue:", np.min(revenue_array))
print("Maximum Revenue:", np.max(revenue_array))
print("Average Revenue:", np.mean(revenue_array))
```

```
Total Revenue: 222366.5
Minimum Revenue: 360.0
Maximum Revenue: 5850.0
Average Revenue: 2223.665
```

```
# Daily operations
daily_revenue = df.groupby("Date")["Revenue"].sum()

print("Daily Total Revenue:", daily_revenue.sum())
print("Minimum Daily Revenue:", daily_revenue.min())
print("Maximum Daily Revenue:", daily_revenue.max())
print("Average Daily Revenue:", daily_revenue.mean())

print("\nDaily Revenue:")
print(daily_revenue)
```

```
Daily Total Revenue: 222366.5
Minimum Daily Revenue: 360.0
Maximum Daily Revenue: 5850.0
Average Daily Revenue: 2223.665

Daily Revenue:
Date
2026-01-01    1615.0
2026-01-02    1140.0
2026-01-03    2280.0
2026-01-04     960.0
2026-01-05    1120.0
               ...
2026-04-06    1728.0
2026-04-07    1188.0
2026-04-08    3600.0
2026-04-09    4200.0
```

```
2026-04-10    1360.0
Name: Revenue, Length: 100, dtype: float64
```

```
#Data exploration
print("First 5 records:")
print(df.head())

print("\nLast 5 records:")
print(df.tail())

print("\nDataset shape:")
print(df.shape)

print("\nColumn names:")
print(df.columns)

print("\nDataset information:")
df.info()

print("\nStatistical summary:")
print(df.describe())

print("\nDuplicate records:", df.duplicated().sum())

print("\nMissing values:")
print(df.isnull().sum())
```

```
96  2026-04-07       Tea          22          60                10         66
97  2026-04-08    Coffee          30         120                 0         32
98  2026-04-09  Sandwich          28         150                 0         56
99  2026-04-10    Muffin          20          80                15         29

    Revenue
95   1728.0
96   1188.0
97   3600.0
98   4200.0
99   1360.0

Dataset shape:
(100, 7)

Column names:
Index(['Date', 'Product', 'Units_Sold', 'Unit_Price', 'Discount_Percent',
       'Customers', 'Revenue'],
      dtype='object')

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Date              100 non-null    object
 1   Product           100 non-null    object
 2   Units_Sold        100 non-null    int64
 3   Unit_Price        100 non-null    int64
 4   Discount_Percent  100 non-null    int64
 5   Customers         100 non-null    int64
 6   Revenue           100 non-null    float64
dtypes: float64(1), int64(4), object(2)
memory usage: 5.6+ KB
```

```
Unit_Price          0
Discount_Percent    0
Customers           0
Revenue             0
dtype: int64
```

```
# creating new col
df["Revenue_Per_Customer"] = (
    df["Revenue"] / df["Customers"]
).round(2)

print(df[["Date", "Revenue", "Customers",
          "Revenue_Per_Customer"]].head())
```

```
         Date  Revenue  Customers  Revenue_Per_Customer
0  2026-01-01   1615.0         26                 62.12
1  2026-01-02   1140.0         52                 21.92
2  2026-01-03   2280.0         98                 23.27
3  2026-01-04    960.0         44                 21.82
4  2026-01-05   1120.0         30                 37.33
```

```
#performing operations on the new col
new_array = np.array(df["Revenue_Per_Customer"])

print("Total:", np.sum(new_array))
print("Minimum:", np.min(new_array))
print("Average:", np.mean(new_array))
```

```
Total: 4262.629999999999
Minimum: 5.47
Average: 42.62629999999999
```

```
#filtering - Days with revenue per customer greater than 30
high_revenue = df[df["Revenue_Per_Customer"] > 30]

print(high_revenue)
print("Number of records:", len(high_revenue))
```

```
          Date   Product  Units_Sold  Unit_Price  Discount_Percent  Customers  \
0   2026-01-01     Juice          17         100                 5         26
4   2026-01-05    Muffin          14          80                 0         30
5   2026-01-06  Sandwich          19         150                 0         59
6   2026-01-07    Coffee          40         120                 5         61
7   2026-01-08  Sandwich          35         150                 0         73
10  2026-01-11  Sandwich          34         150                 5         62
12  2026-01-13     Juice          35         100                15         97
14  2026-01-15       Tea          38          60                 0         60
15  2026-01-16     Juice          15         100                 5         22
19  2026-01-20  Sandwich          25         150                10         71
21  2026-01-22  Sandwich          33         150                 5         28
22  2026-01-23    Muffin          40          80                 5         78
23  2026-01-24    Coffee          28         120                15         31
24  2026-01-25  Sandwich          19         150                 5         80
25  2026-01-26  Sandwich          19         150                 5         53
26  2026-01-27     Juice          20         100                 0         20
30  2026-01-31    Muffin          17          80                 0         34
34  2026-02-04    Coffee          32         120                 5         54
35  2026-02-05     Juice          31         100                 0         57
36  2026-02-06    Coffee          30         120                 5         71
37  2026-02-07    Coffee          21         120                 0         83
38  2026-02-08  Sandwich          30         150                 5         88
39  2026-02-09     Juice          10         100                10         21
41  2026-02-11    Coffee          23         120                 0         28
42  2026-02-12     Juice          38         100                15         43
44  2026-02-14    Muffin          22          80                 0         40
46  2026-02-16    Coffee          22         120                 0         57
47  2026-02-17     Juice          21         100                10         38
52  2026-02-22    Coffee          29         120                 5         95
53  2026-02-23  Sandwich          27         150                 0         90
56  2026-02-26  Sandwich          39         150                 0         43
58  2026-02-28    Muffin          17          80                15         29
61  2026-03-03    Coffee          39         120                10         54
62  2026-03-04    Muffin          18          80                10         29
63  2026-03-05    Coffee          37         120                 5         72
66  2026-03-08  Sandwich          21         150                 5         42
68  2026-03-10  Sandwich          32         150                 0         63
69  2026-03-11       Tea          39          60                10         40
70  2026-03-12     Juice          14         100                 0         41
```

```
71  2026-03-13  Sandwich          33         150                15         39
73  2026-03-15     Juice          30         100                 0         80
74  2026-03-16    Coffee          33         120                10         88
76  2026-03-18     Juice          31         100                 0         97
77  2026-03-19    Muffin          14          80                 5         28
79  2026-03-21     Juice           8         100                 0         25
80  2026-03-22     Juice          21         100                 0         58
83  2026-03-25       Tea          21          60                 5         31
84  2026-03-26    Coffee          30         120                 0         35
86  2026-03-28  Sandwich          30         150                 0         45
87  2026-03-29     Juice          16         100                 0         45
88  2026-03-30    Coffee          34         120                 5         94
89  2026-03-31     Juice          25         100                 0         31
90  2026-04-01  Sandwich          24         150                 5         94
93  2026-04-04     Juice          35         100                 0         33
95  2026-04-06       Tea          32          60                10         43
97  2026-04-08    Coffee          30         120                 0         32
```

```
#filtering - Days with revenue per customer less than or equal to 30
low_revenue = df[df["Revenue_Per_Customer"] <= 30]

print(low_revenue)
print("Number of records:", len(low_revenue))
```

```
          Date   Product  Units_Sold  Unit_Price  Discount_Percent  Customers  \
1   2026-01-02       Tea          20          60                 5         52
2   2026-01-03       Tea          40          60                 5         98
3   2026-01-04    Muffin          12          80                 0         44
8   2026-01-09  Sandwich           6         150                 0         45
9   2026-01-10       Tea          13          60                 0         97
11  2026-01-12    Coffee           7         120                 0         43
13  2026-01-14  Sandwich          15         150                 5         94
16  2026-01-17       Tea          20          60                 0         68
17  2026-01-18       Tea          28          60                15         64
18  2026-01-19     Juice           9         100                 5         32
20  2026-01-21     Juice          23         100                 0         88
27  2026-01-28       Tea          34          60                 0         98
28  2026-01-29       Tea          16          60                 5         33
29  2026-01-30    Muffin          11          80                 0         68
31  2026-02-01    Coffee           5         120                10         95
32  2026-02-02       Tea           8          60                10         79
33  2026-02-03       Tea           8          60                 0         85
40  2026-02-10       Tea          37          60                10         73
43  2026-02-13    Coffee          10         120                 0         87
45  2026-02-15       Tea          30          60                 5         84
48  2026-02-18    Muffin          10          80                10         47
49  2026-02-19    Muffin          18          80                 0         62
50  2026-02-20    Muffin          13          80                15         81
51  2026-02-21    Muffin          15          80                 0         69
54  2026-02-24    Muffin          26          80                15         81
55  2026-02-25    Muffin          18          80                 0         68
57  2026-02-27    Coffee           8         120                 0         53
59  2026-03-01       Tea           9          60                10         50
60  2026-03-02     Juice          17         100                10         95
64  2026-03-06    Coffee          22         120                 0         97
65  2026-03-07     Juice          30         100                10         90
67  2026-03-09       Tea          14          60                 0         56
72  2026-03-14  Sandwich          14         150                 0        100
75  2026-03-17    Muffin          21          80                 0         86
78  2026-03-20    Muffin           7          80                10         91
81  2026-03-23       Tea          37          60                10         68
82  2026-03-24    Coffee           9         120                15         48
85  2026-03-27       Tea          12          60                 0         86
91  2026-04-02  Sandwich          11         150                 5         94
92  2026-04-03    Muffin          21          80                 5         68
94  2026-04-05    Muffin           5          80                10         32
96  2026-04-07       Tea          22          60                10         66

    Revenue  Revenue_Per_Customer
1    1140.0                21.92
2    2280.0                23.27
3     960.0                21.82
8     900.0                20.00
9     780.0                 8.04
11    840.0                19.53
13   2137.5                22.74
16   1200.0                17.65
17   1428.0                22.31
18    855.0                26.72
20   2300.0                26.14
27   2040.0                20.82
28    912.0                27.64
```

```
#Save the updated dataset
df.to_csv("coffee_shop_sales_analyzed.csv", index=False)

files.download("coffee_shop_sales_analyzed.csv")
```